

Digital Research Toolkit for Linguists

Week 8: Data visualization and exporting

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May 27, 2025

Psycholinguistics and Cognitive Modeling Lab

Homework

Homework

Name your files with your name and assignment number.

Error: object 'moses_clean' not found

→ remember to load the data

```
mosess <- read.csv("moses_clean.csv")
```

```
moses_clean.csv <- read.csv("moses_clean.csv")
```

→ what would be a better name?

You need to load the packages only once.

ggplot2 typically expects **one row per observation** for plotting.

→ remember to group and/or summarize your data

**⚠ Using the `magrittr` pipe in your homework will result in
»failing the assignment«.**

If you wish to use the `magrittr` pipe in your homework, you need to submit a 5 A4 page explanation why you needed to do so. **⚠**

POUR bingo

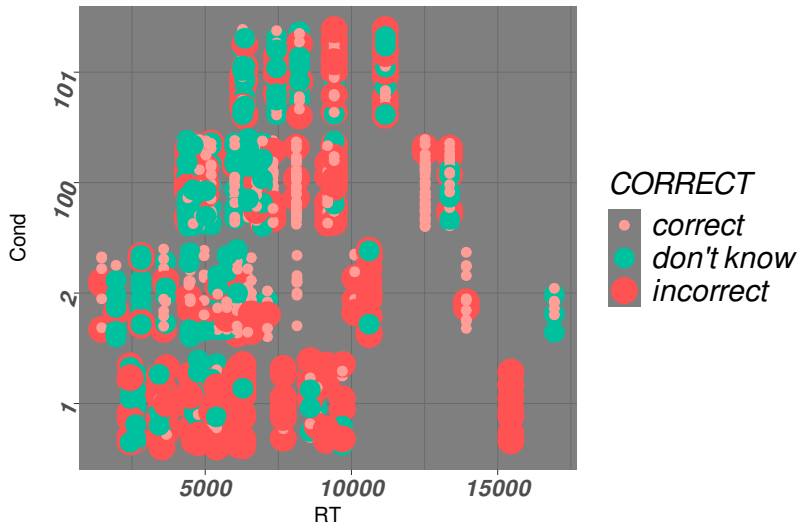


Table of contents

1. Colors
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Colors

Color palette types

Discrete

Uses distinct, unrelated colors for categorical data.



Divergent

Shows data with a central midpoint, using contrasting colors to highlight deviations.



Sequential

Represents data with a gradient of colors to indicate ordered values.



Adobe color.adobe.com

Coolers coolers.co

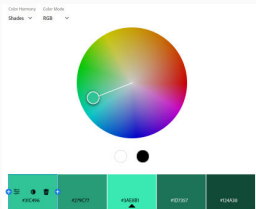
LearnUI www.learnui.design

Test the colors for accessibility and BW print:

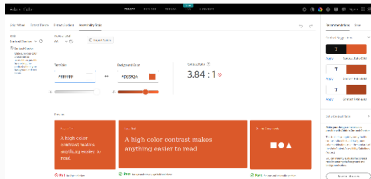
webaim.org/resources/contrastchecker

```
my.palette <- c("#8c510a", "#d8b365", "#f6e8c3", "#f5f5f5")  
plot + scale_color_manual(values = my.palette)  
  
plot + scale_fill_manual(values =  
  c("#8c510a", "#d8b365", "#f6e8c3", "#f5f5f5"))
```

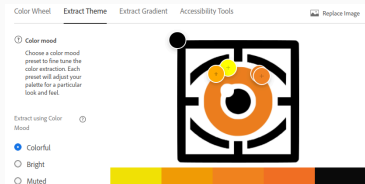

Assemble colors



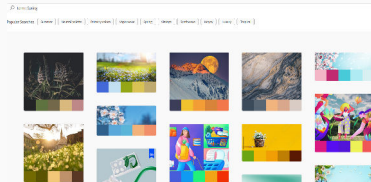
Check contrast



Extract from image

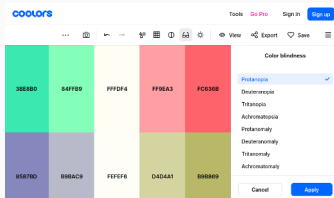


Browse trends

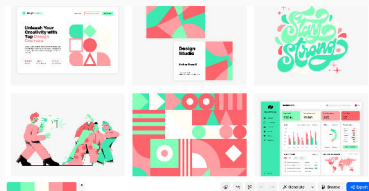


Coolors

Assemble colors and check for color blindness



Visualize the palette



Browse trends



Generate color palettes

PALETTE SINGLE HUE DIVERGENT

PALETTE GENERATOR

NUMBER OF COLORS: 3 4 5 6 7 8

BACKGROUND COLOR: LIGHT DARK



#003f5c #374c80 #7a5195 #bc5090 #e15675 #f2764a #ffa600

ACTIONS: COPY HEX VALUES EXPORT AS SVG

IN CONTEXT



Check contrast

Show me the closest variations of  #4ac4e2

that contrast against the color  #ffffff

enough to meet AA Guidelines

RESULTS

FOR LARGE/BOLD TEXT

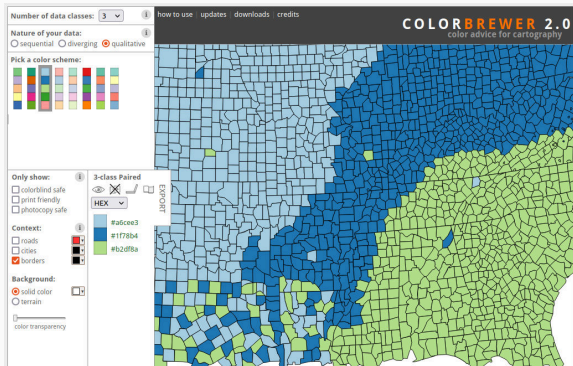
Try this combo instead:



FOR SMALL TEXT

Try this combo instead:





There's a package for that

RColorBrewer

```
scale_fill_brewer(..., type="div", palette=1)
```

viridis

```
scale_colour_viridis_c(..., option="inferno")
```

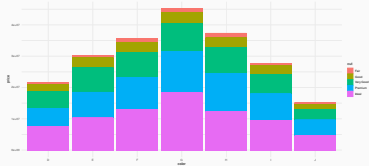
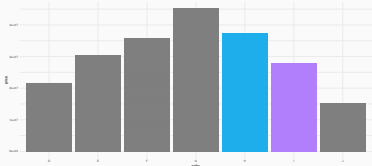
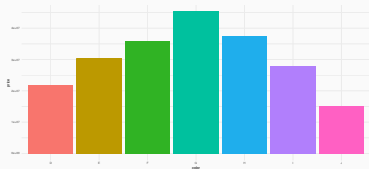
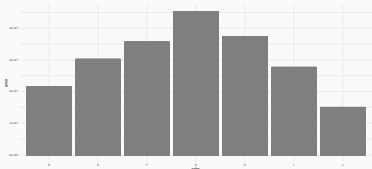
wesanderson

```
scale_fill_manual(values=wes_palette("BottleRocket1"))
```

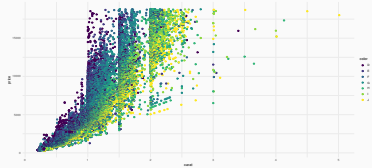
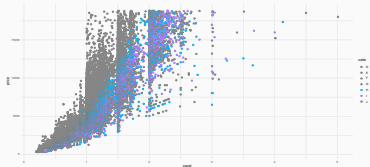
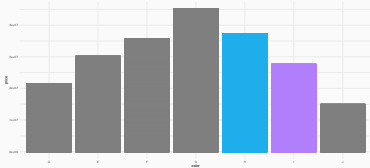
Others, e.g. ggsci (for journals)

Color use guidelines

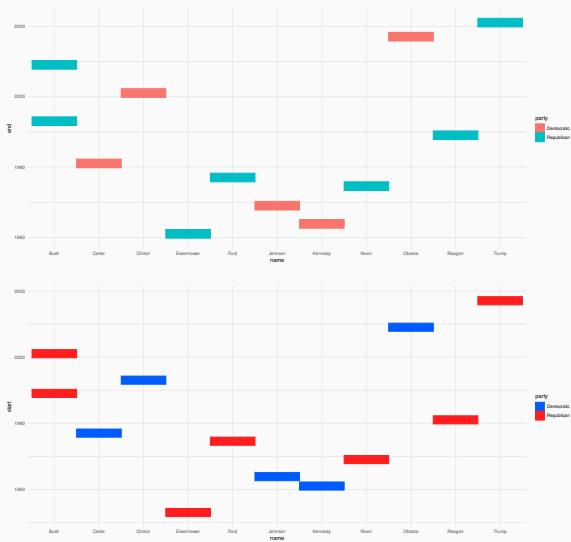
Avoid unnecessary usage of color



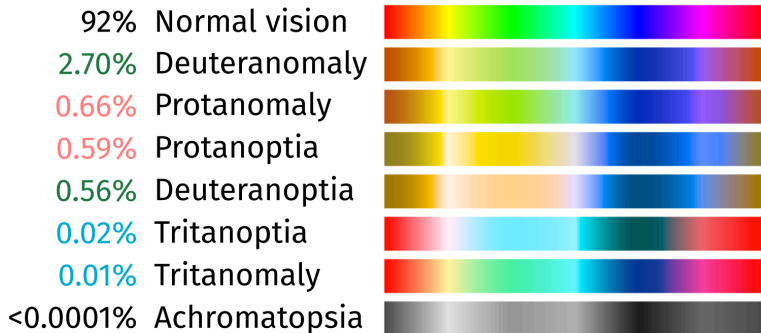
Be consistent



Use in a meaningful way

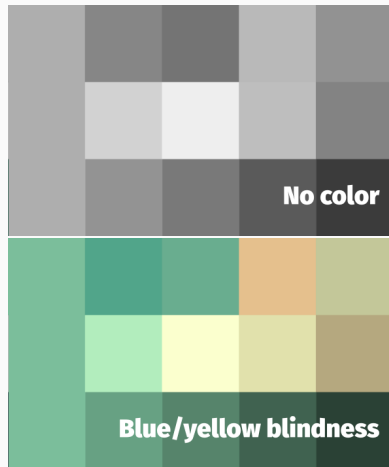


Pay attention to color blindness



1-2 people in this class

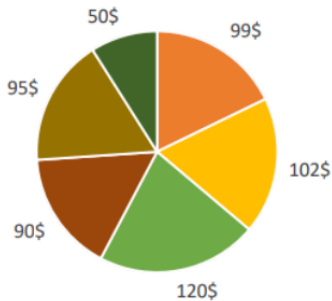
Perceivable colors



Lying with plots

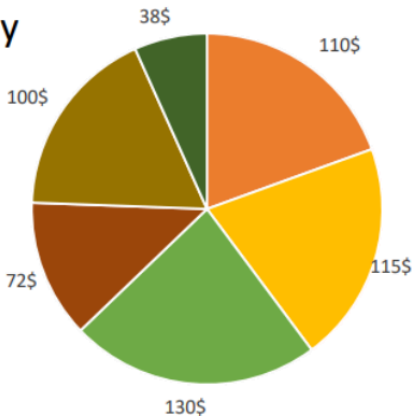
Pie plots: Manipulate the size

Total Sales by Company
(Millions)



Last year

Apple Amazon Google Microsoft Facebook Volkswagen



This year

Fallenbüchel (2019)

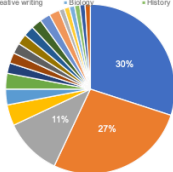
Pie plots: Manipulate categories

Too many slices

Declared majors

% OF ANDERSON HALL RESIDENTS

- Undecided
- Education
- Accounting
- Management
- Mathematics
- Creative writing
- Biology
- Political science
- Economics
- Biochemistry
- Forensics
- Biology
- Finance
- English
- Marketing
- Computer science
- Geology
- History

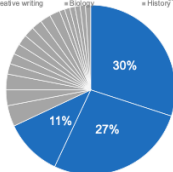


De-emphasize

Declared majors

% OF ANDERSON HALL RESIDENTS

- Undecided
- Education
- Accounting
- Management
- Mathematics
- Creative writing
- Biology
- Political science
- Economics
- Biochemistry
- Forensics
- Biology
- Finance
- English
- Marketing
- Computer science
- Geology
- History

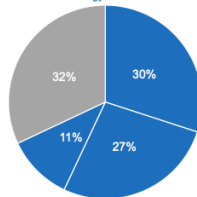


Aggregate

Declared majors

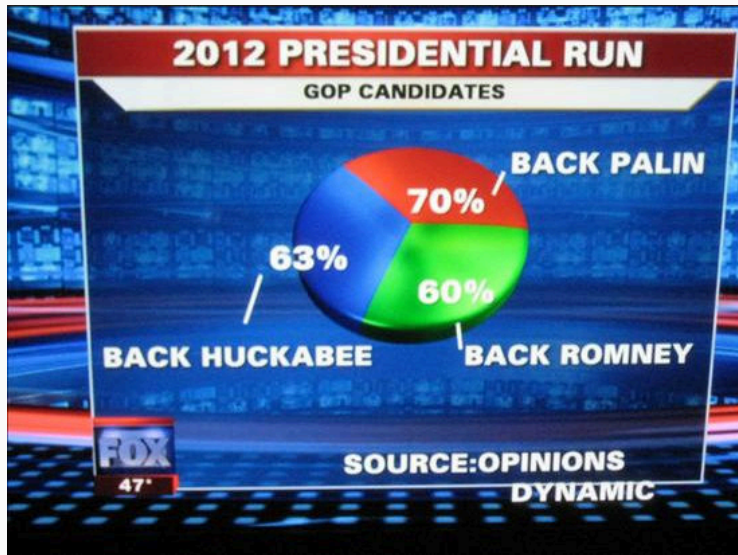
% OF ANDERSON HALL RESIDENTS

- Undecided
- Biology
- Finance
- All other (17)

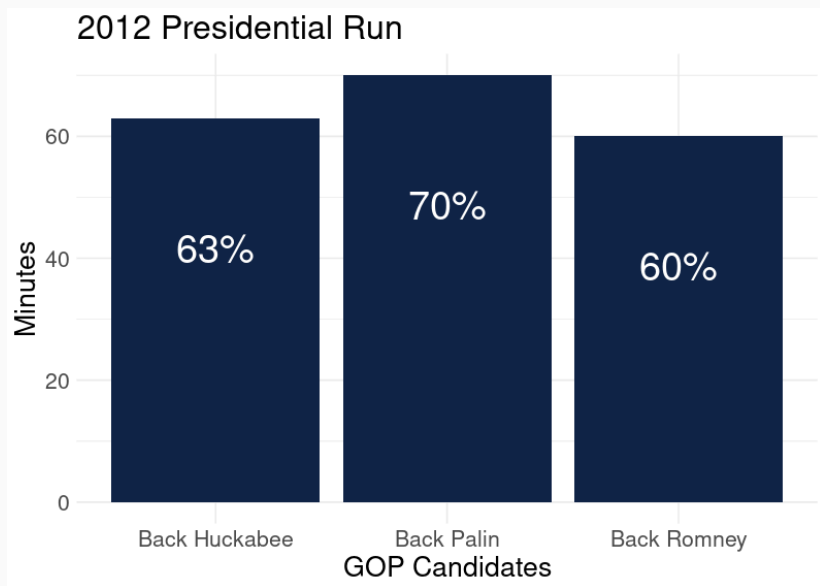


Ricks (2020)

Pie plots: Wrong plot type



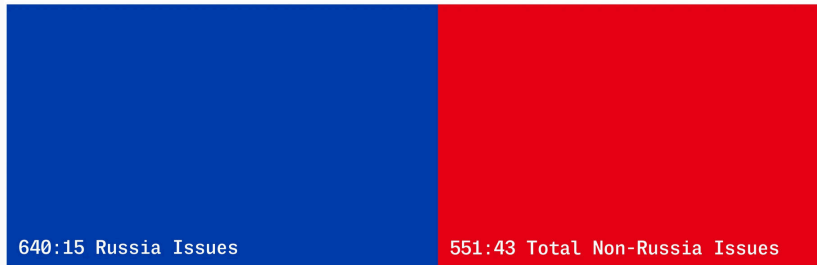
Pie plots: Wrong plot type



RUSSIA ISSUES VS. NON-RUSSIA ISSUES

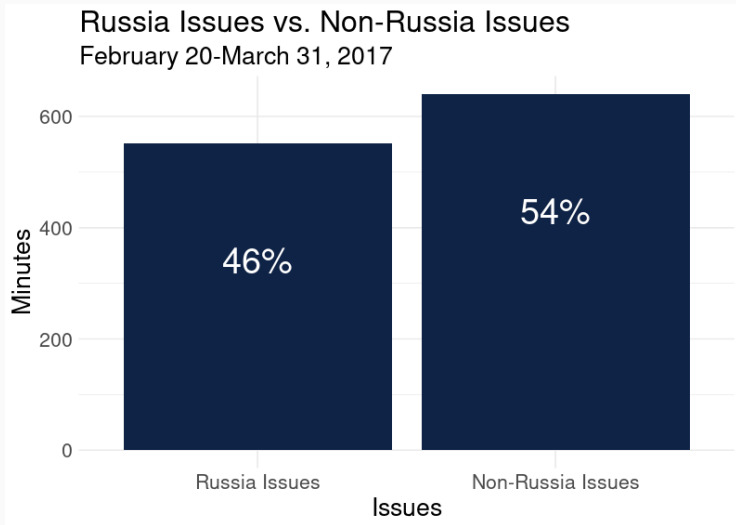
February 20–March 31, 2017

1191:58 (min:sec) Total Show Minutes

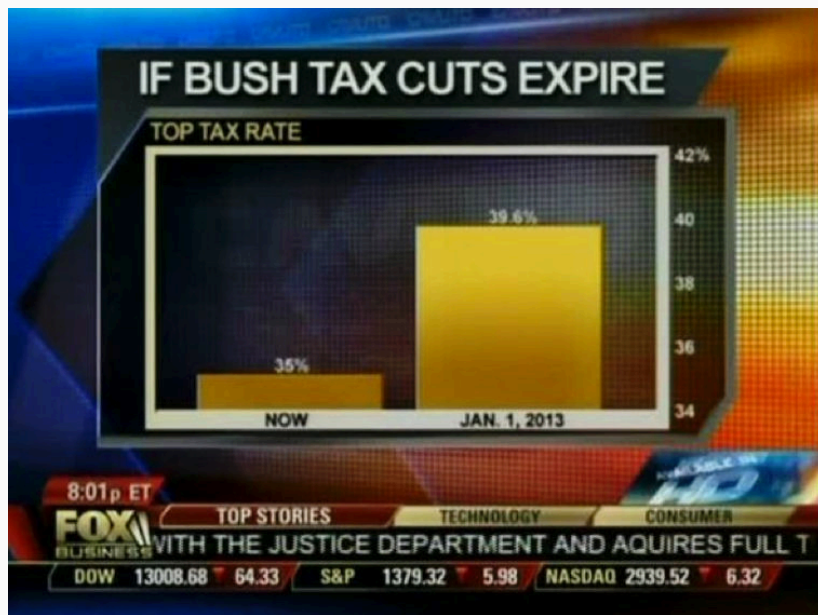


Maté (2017)

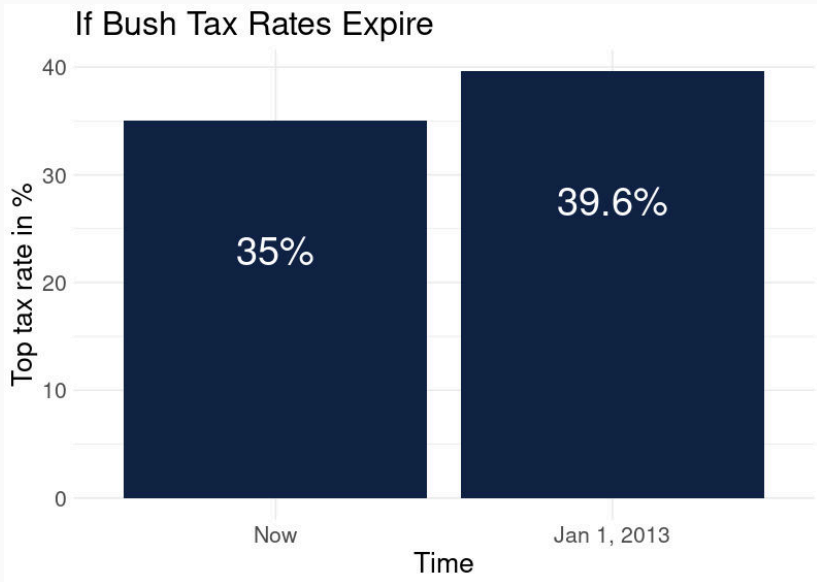
Bar plots: Stack (+ color)



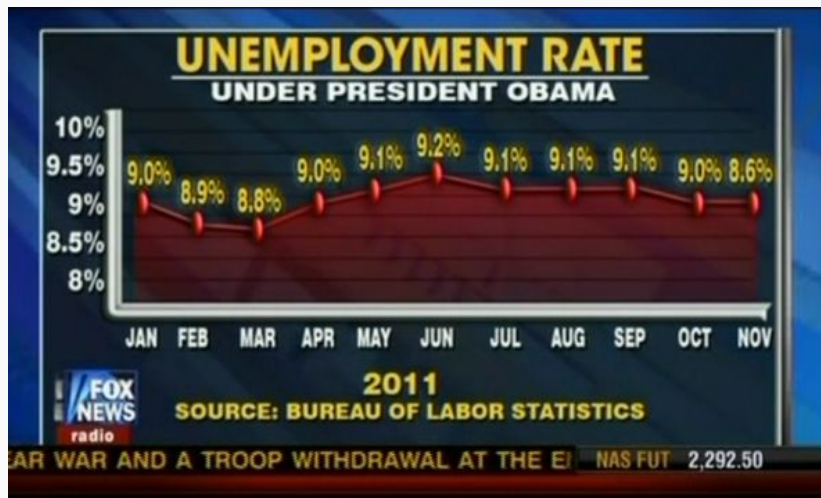
X/Y plots: Omit the baseline



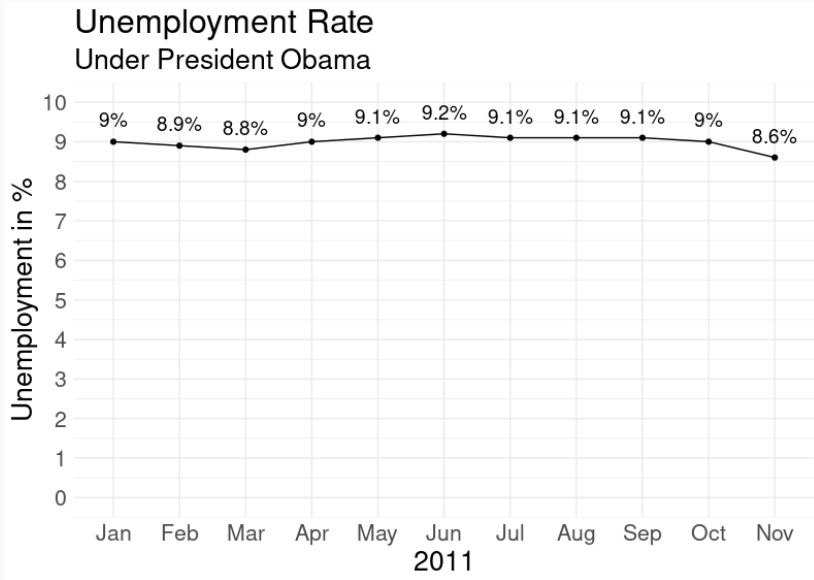
X/Y plots: Omit the baseline



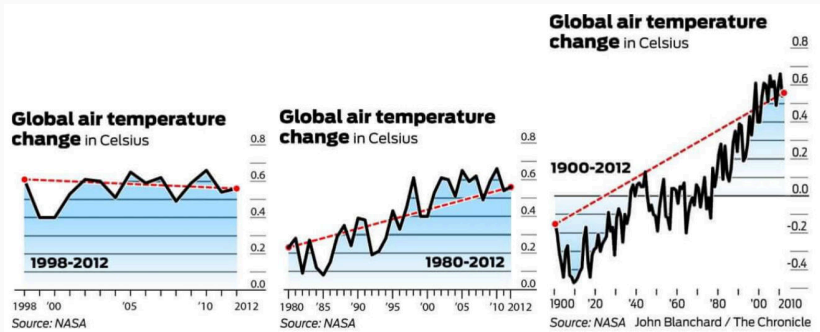
X/Y plots: Manipulate the Y-axis



X/Y plots: Manipulate the Y-axis

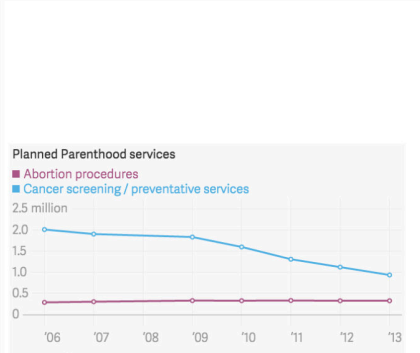
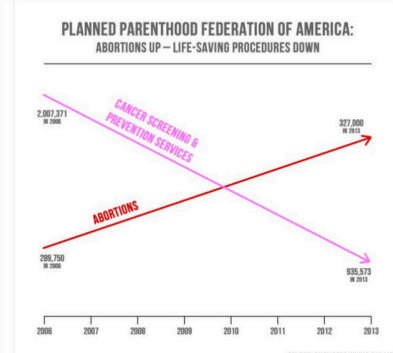


X/Y plots: Pick the range



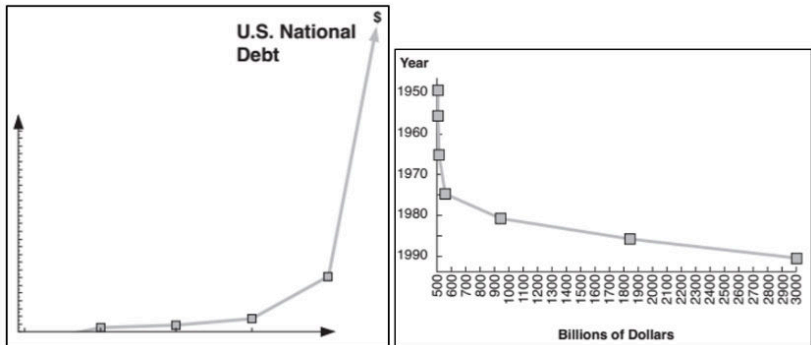
Fallenbüchel (2019)

X/Y plots: Fuck the y-axis entirely



Fallenbüchel (2019)

X/Y plots: Flip plot

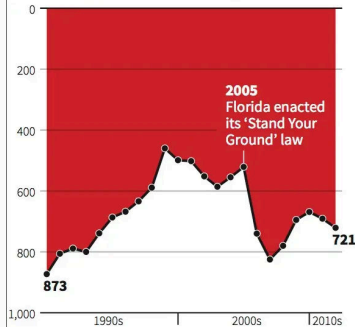


Fallenbüchel (2019)

X/Y plots: Flip plot

Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

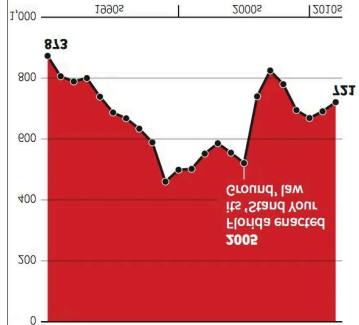
C. Chan 16/02/2014

REUTERS

C. Chan 16/02/2014

REUTERS

Source: Florida Department of Law Enforcement



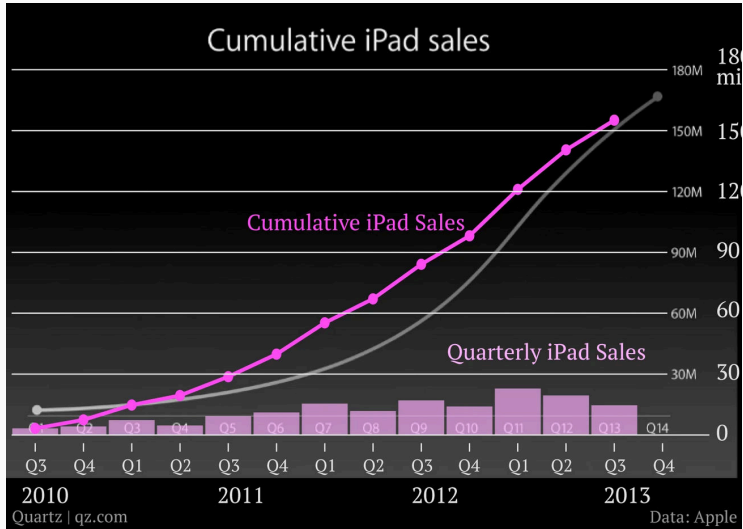
Number of murders committed using firearms

Gun deaths in Florida

X/Y plots: Cumulative growth



X/Y plots: Cumulative growth



Yanofsky (2013)

X/Y plots: Cherrypick data



realdonaldtrump • Follow

realdonaldtrump Thank you very much, working hard!

Load more comments

riot_racer @figboot31977 I know your a Donald trump supporter. Kind of obvious

figboot31977 @riot_racer Here's a thought for ya.... You don't know crap!!! GET a LIFE and STAY the #&# OUT OF MINE!!!

riot_racer @figboot31977 I'm not in yours. If I was then I'd have a bigger understanding of who you are. Like I said if you support



625,072 likes

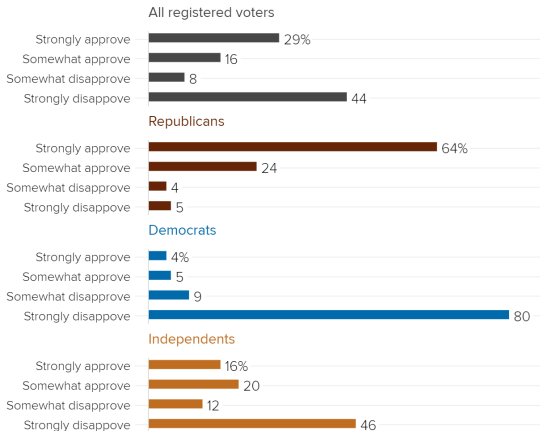
JULY 26

Log in to like or comment.



X/Y plots: Cherrypick data

Strength of Trump approval/disapproval by party



NBC NEWS

Data: NBC News/Wall Street Journal poll. July 15-18, 2018.

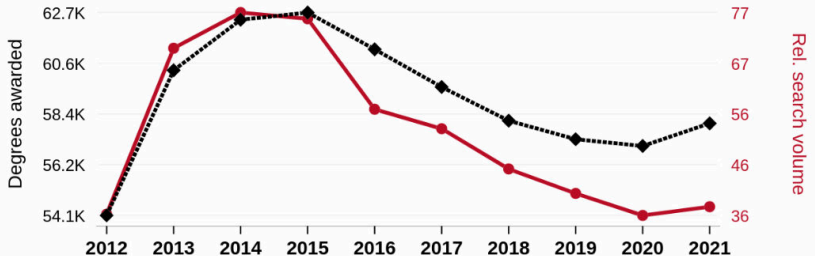
Murray (2018)

X/Y plots: Correlate axes

Bachelor's degrees awarded in law enforcement

correlates with

Google searches for 'sleepwalking'



◆ Bachelor's degrees conferred by postsecondary institutions, in field of study: Homeland security, law enforcement, and firefighting · Source: National Center for Education Statistics

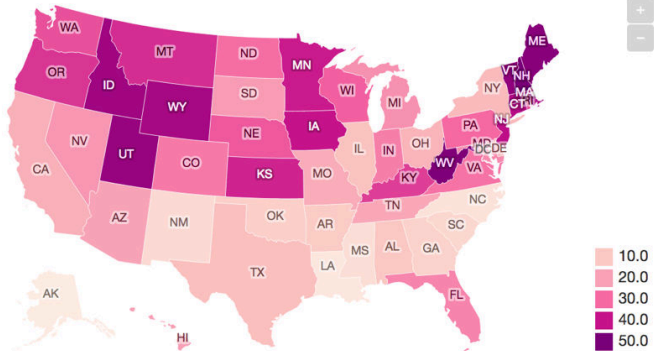
● Relative volume of Google searches for 'sleepwalking' (Worldwide, without quotes) · Source: Google Trends

2012-2021, $r=0.903$, $r^2=0.815$, $p<0.01$ · tylervigen.com/spurious/correlation/1532

Vigen (2024)

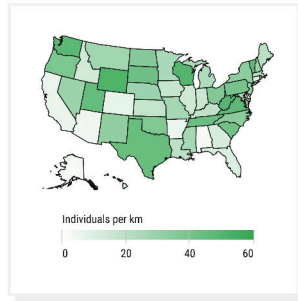
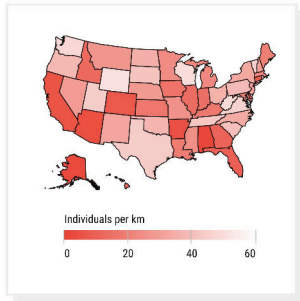
Colors: go against convention

Which states have the most STIs?



McCready (2020)

Colors: go against convention



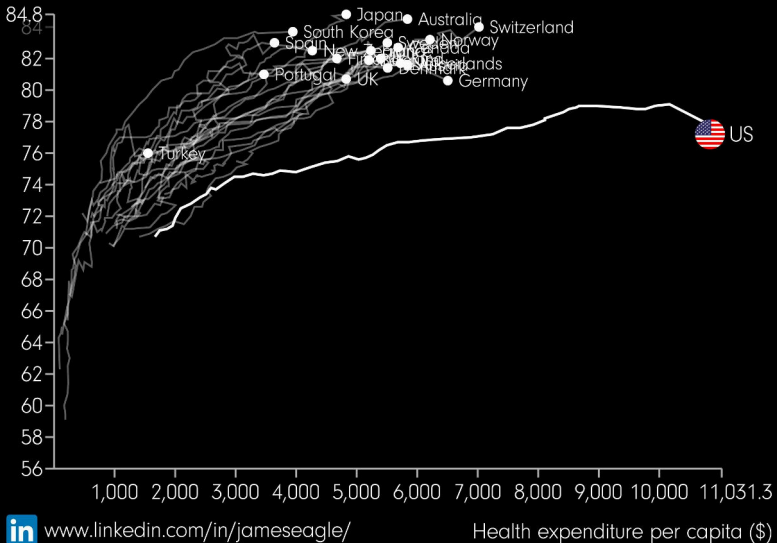
McCready (2020)

Informing with plots

Life expectancy vs. health expenditure

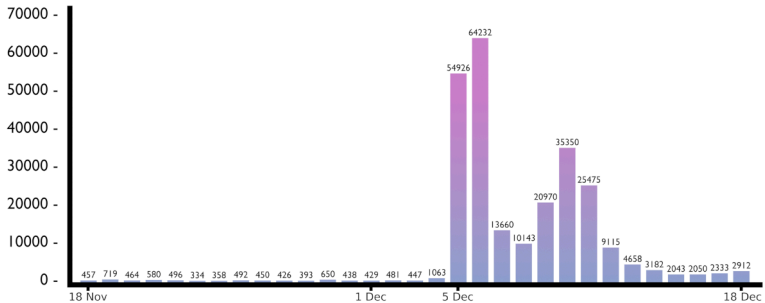
Life expectancy (years)

2021

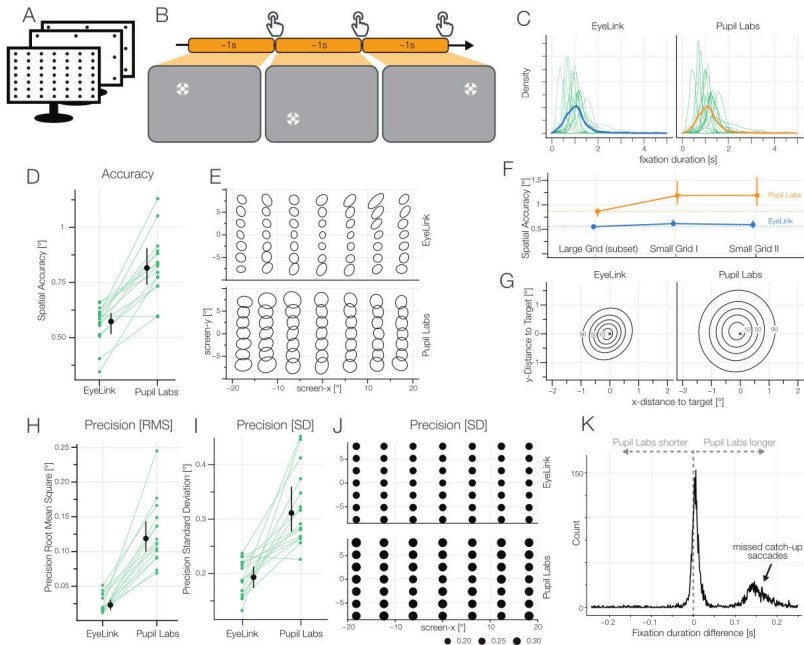


Views for the "Jury Nullification" Wikipedia page

(Nov - Dec 2024)

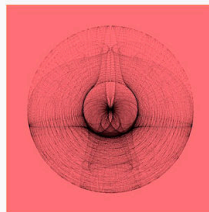
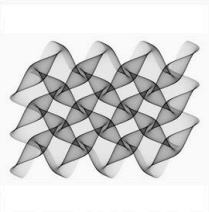
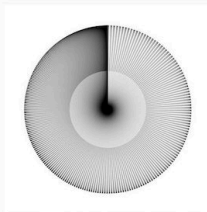
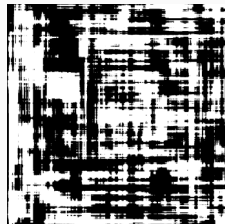


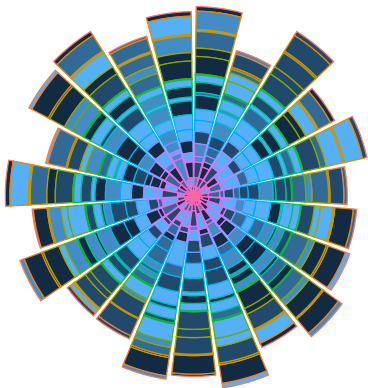
Source: Pageview Analysis



Generative art

Start out with `aRtsy` and `generativeart`, but also a mix of `ggplot2`, `ggforce`, `rayshader`, `ggpattern`, `gridExtra`



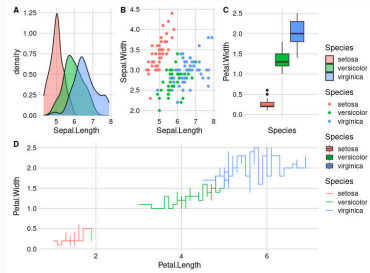


Moving on from R

Arranging plots

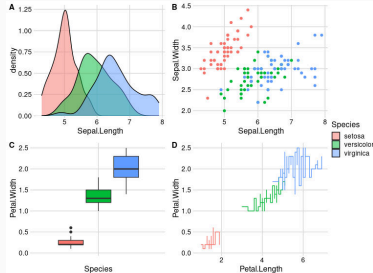
patchwork

makes it ridiculously simple to combine separate ggplots into the same graphic.



cowplot

provides various features to make plots beautiful, including aligning and arranging plots.



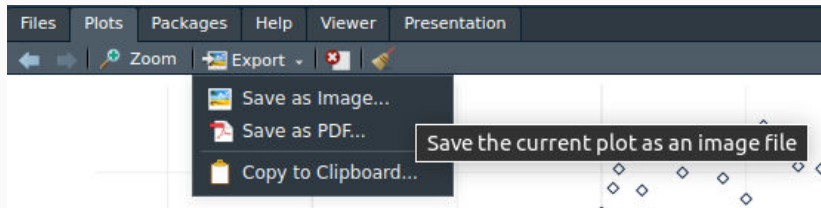
Exporting plots

```
ggsave("FILENAME", width=NR, height=NR, dpi =300)
```

Saves the last plot by default, but you can specify `plot = my.plot`

A common standard for high-quality prints is 300 DPI (dots per inch)

Choose PDF or PNG file extension for printing, SVG for more editing.



Exporting dataframes

Save a data frame to the **current working directory**.

<code>write_csv(WHAT, "NAME")</code>	comma
<code>write_tsv()</code>	tab
<code>write_excel_csv()</code>	CSV for Excel
<code>write_delim()</code>	specify how to separate

```
write_csv(moses_accuracy, "Moses accuracy.csv")  
write_delim(moses_accuracy, "Moses accuracy.txt",  
            delim = ":")
```

Questions?

Wrap-up

Summary

- ✓ R programming basics and RStudio IDE
- ✓ write scripts
- ✓ file encoding, variable naming, and tidy code with pipes
- ✓ install and load packages
- ✓ import/export data from/to the working directory
- ✓ save and remove objects in the environment
- ✓ preprocess raw data (filtering, renaming, arranging, mutating, selecting, if else)
- ✓ make sense of data (merging, grouping, summarizing)
- ✓ print and visualize the results
- ✓ find and ask for help
- documentation

Homework assignment due June 2nd 12:00

- ❓ Complete assignment 8 (→ ILIAS)
- ❓ Resubmit an ugly plot (if you want to).
- ❓ Install Quarto: <https://quarto.org/docs/get-started/>
- ❓ Watch the Quarto introductory video: https://youtu.be/_f3latm0hew?si=xxovQvYkUosC_4uB

Next time: Reporting on data



References



Ehinger, Benedikt V et al. (2019). **“A new comprehensive eye-tracking test battery concurrently evaluating the Pupil Labs glasses and the EyeLink 1000”**. In: *PeerJ* 7, e7086. DOI:

10.7717/peerj.7086. URL:

<https://doi.org/10.7717/peerj.7086>.



Fallenbüchel, Florian (2019). ***How To Lie With Charts: Elaboration on the presentation for the seminar “How do I lie with statistics?”***

Report. URL: https://hci.iwr.uni-heidelberg.de/system/files/private/downloads/121410023/florian-fallenbuechel_report.pdf.



Maté, Aaron (Apr. 2017). *MSNBC's Rachel Maddow Sees a "Russia Connection" Lurking Around Every Corner*. URL: <https://theintercept.com/2017/04/12/msnbcs-rachel-maddow-sees-a-russia-connection-lurking-around-every-corner/> (visited on 05/23/2024).



McCready, Ryan (2020). *How to Avoid Misleading Graphs: Practical Tips and Examples*. URL: <https://venngage.com/blog/misleading-graphs/> (visited on 05/23/2024).



Murray, Mark (2018). *NBC/WSJ poll: Public gives Trump thumbs down on Russia, thumbs up on economy*. URL: <https://www.nbcnews.com/politics/first-read/nbc-wsj-poll-public-gives-trump-thumbs-down-russia-thumbs-n893266> (visited on 05/23/2024).



Ricks, Elizabeth (May 2020). *What Is a Pie Chart?* URL: <https://www.storytellingwithdata.com/blog/2020/5/14/what-is-a-pie-chart> (visited on 05/23/2024).



Vigen, Tyler (2024). *Spurious Correlations*. URL:

<https://tylervigen.com/spurious-correlations> (visited on 05/23/2024).



Yanofsky, David (2013). *Apple is either terrible at designing charts, or thinks you won't notice the difference*. URL:

<https://qz.com/138458/apple-is-either-terrible-at-designing-charts-or-thinks-you-wont-notice-the-difference> (visited on 05/23/2024).